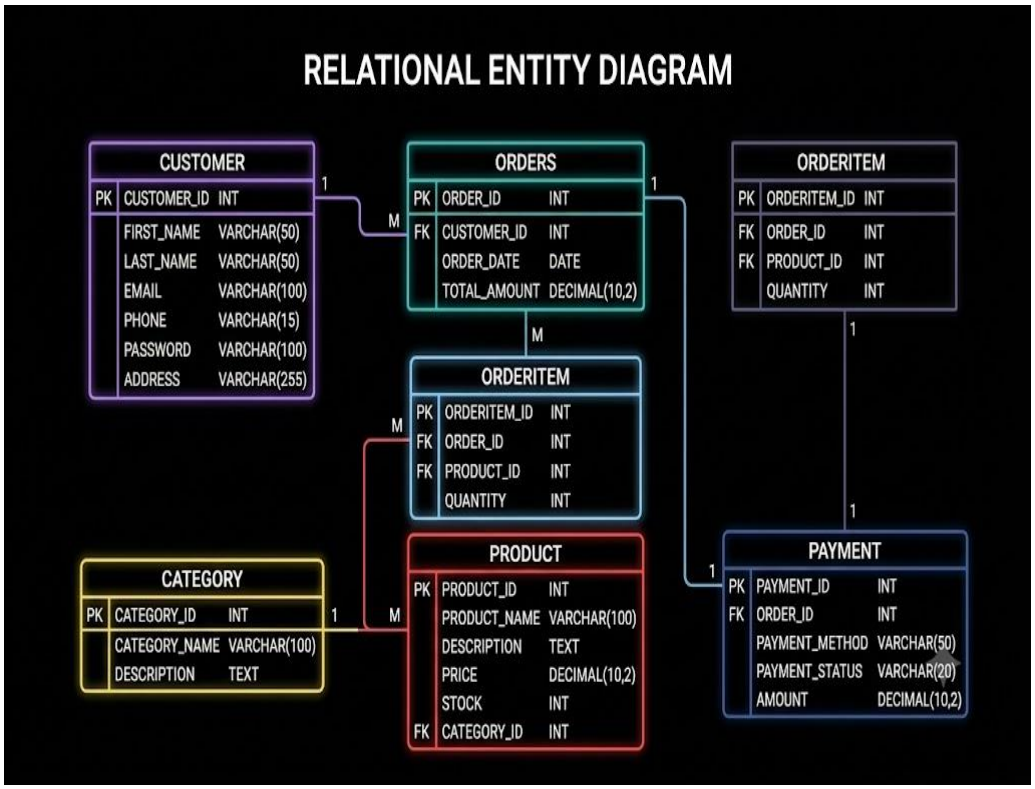
DATABASE RELATIONSHIP ANALYSIS USING JOINS

TATA CLiQ E-Commerce Database Management System

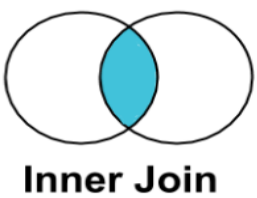
1.Combining Multiple Tables

The ER Diagram represents the entities and relationships in the TATA CLiQ E-Commerce Database, including Customer, Product, Orders, and Payment. It shows how tables are connected using Primary Keys and Foreign Keys to maintain data consistency.



2.INNER JOIN

INNER JOIN retrieves matching records from two or more tables based on a related column



2.1 Customer And Orders

INNER JOIN retrieves matching customer and order details using Customer_ID from both tables

```
SQL> SELECT o.order_id, o.order_date,
  2         p.payment_method, p.payment_status,
  3         p.amount
  4  FROM orders o
  5  INNER JOIN payment p
  6  ON o.order_id = p.order_id;

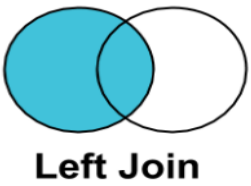
  ORDER_ID ORDER_DAT PAYMENT_METHOD       PAYMENT_STATUS          AMOUNT
---------- --------- -------------------- -------------------- ----------
      1001 20-SEP-26 UPI                  Success                    2500
      1002 25-SEP-26 Credit Card          Success                    1800
      1003 22-SEP-26 Debit Card           Success                    3200
      1004 23-SEP-26 Cash on Delivery     Success                    1500
      1005 24-SEP-26 UPI                  Success                    4200
      1001 20-SEP-26 UPI                  Success                    2500
      1002 25-SEP-26 Credit Card          Success                    1800
      1003 22-SEP-26 Debit Card           Success                    3200
      1004 23-SEP-26 Cash on Delivery     Success                    1500
      1005 24-SEP-26 UPI                  Success                    4200
      1001 20-SEP-26 UPI                  Success                    2500

  ORDER_ID ORDER_DAT PAYMENT_METHOD       PAYMENT_STATUS          AMOUNT
---------- --------- -------------------- -------------------- ----------
      1002 25-SEP-26 Credit Card          Success                    1800
      1003 22-SEP-26 Debit Card           Failed                     3200
      1004 23-SEP-26 Cash on Delivery     Success                    1500
      1005 24-SEP-26 UPI                  Success                    4200

15 rows selected.
```

3.LEFT JOIN

LEFT JOIN returns all records from the left table and matching records from the right table. If there is no match, the right table's columns return NULL values



3.1 Display All Customers and Their Orders

LEFT JOIN displays all customers and their order details

```
SQL> SELECT c.customer_id,
  2         c.first_name,
  3         c.last_name,
  4         o.order_id,
  5         o.order_date,
  6         o.total_amount
  7  FROM customer c
  8  LEFT JOIN orders o
  9  ON c.customer_id = o.customer_id;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                           ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
TOTAL_AMOUNT
------------
        101 Thansila
Begam                                                 1001 20-SEP-26
        2700

        102 Priya
Devi                                                  1002 25-SEP-26
        3000

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                           ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
TOTAL_AMOUNT
------------

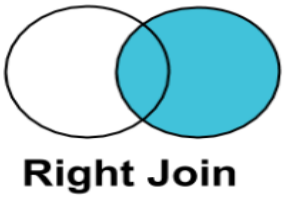
        103 Rahul
Kumar                                                 1003 22-SEP-26
        3200

        101 Thansila
Begam                                                 1004 23-SEP-26

CUSTOMER_ID FIRST_NAME
```

4.RIGHT JOIN

RIGHT JOIN returns all records from the right table and matching records from the left table. If there is no match, the left table's columns return NULL values



4.1 Display All Orders and Customer Details

RIGHT JOIN displays all orders and their matching customer details

```
SQL> SELECT o.order_id,
  2         o.order_date,
  3         o.total_amount,
  4         c.customer_id,
  5         c.first_name,
  6         c.last_name
  7  FROM orders o
  8  RIGHT JOIN customer c
  9  ON o.customer_id = c.customer_id;

  ORDER_ID ORDER_DAT TOTAL_AMOUNT CUSTOMER_ID
---------- --------- ------------ -----------
FIRST_NAME
--------------------------------------------------
LAST_NAME
--------------------------------------------------
      1001 20-SEP-26         2700         101
Thansila
Begam

      1002 25-SEP-26         3000         102
Priya
Devi

  ORDER_ID ORDER_DAT TOTAL_AMOUNT CUSTOMER_ID
---------- --------- ------------ -----------
FIRST_NAME
--------------------------------------------------
LAST_NAME
--------------------------------------------------

      1003 22-SEP-26         3200         103
Rahul
Kumar

      1004 23-SEP-26         1500         101
Thansila

  ORDER_ID ORDER_DAT TOTAL_AMOUNT CUSTOMER_ID
```

5. Complete Order Details

This query retrieves complete order details, including customer information, product details, quantity, order status, and payment information by joining multiple tables

```
SQL> SELECT c.first_name,
  2         c.last_name,
  3         o.order_id,
  4         p.product_name,
  5         oi.quantity,
  6         pay.payment_method,
  7         pay.amount
  8  FROM customer c
  9  JOIN orders o
 10  ON c.customer_id = o.customer_id
 11  JOIN orderitem oi
 12  ON o.order_id = oi.order_id
 13  JOIN product p
 14  ON oi.product_id = p.product_id
 15  JOIN payment pay
 16  ON o.order_id = pay.order_id;

FIRST_NAME
--------------------------------------------
LAST_NAME                                          ORDER_ID
------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------------------------
----
  QUANTITY PAYMENT_METHOD          AMOUNT
---------- -------------------- ----------
Thansila
Begam                                            1001
Samsung Galaxy S24
         2 UPI                        2500
```

6. Customer Purchase History

```
SQL> SELECT c.first_name,
  2         c.last_name,
  3         p.product_name,
  4         o.order_date,
  5         o.total_amount
  6  FROM customer c
  7  JOIN orders o
  8  ON c.customer_id = o.customer_id
  9  JOIN orderitem oi
 10  ON o.order_id = oi.order_id
 11  JOIN product p
 12  ON oi.product_id = p.product_id
 13  WHERE c.customer_id = 101;

FIRST_NAME
--------------------------------------------
LAST_NAME
--------------------------------------------
PRODUCT_NAME
--------------------------------------------
----
ORDER_DAT TOTAL_AMOUNT
--------- ------------
Thansila
Begam
Samsung Galaxy S24
20-SEP-26         2700
```

7. Customer Purchase History with Payment

This query displays customer purchase history with product details, order amounts, and payment methods and statuses

```
SQL> SELECT c.first_name,
  2         c.last_name,
  3         p.product_name,
  4         o.order_date,
  5         o.total_amount,
  6         pay.payment_method,
  7         pay.payment_status
  8  FROM customer c
  9  JOIN orders o
 10  ON c.customer_id = o.customer_id
 11  JOIN orderitem oi
 12  ON o.order_id = oi.order_id
 13  JOIN product p
 14  ON oi.product_id = p.product_id
 15  JOIN payment pay
 16  ON o.order_id = pay.order_id
 17  WHERE c.customer_id = 101;

FIRST_NAME
--------------------------------------------------
LAST_NAME
--------------------------------------------------
PRODUCT_NAME
--------------------------------------------------------------------------------
----
ORDER_DAT TOTAL_AMOUNT PAYMENT_METHOD        PAYMENT_STATUS
--------- ------------ --------------------- ---------------------
Thansila
Begam
Samsung Galaxy S24
20-SEP-26          2700 UPI                   Success
```

8. Multi-Table Business Report

```
SQL> SELECT c.customer_id,
  2         c.first_name,
  3         c.last_name,
  4         COUNT(o.order_id) AS total_orders,
  5         SUM(o.total_amount) AS total_spent
  6  FROM customer c
  7  JOIN orders o
  8  ON c.customer_id = o.customer_id
  9  GROUP BY c.customer_id,
 10           c.first_name,
 11           c.last_name
 12  ORDER BY total_spent DESC;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          TOTAL_ORDERS TOTAL_SPENT
-------------------------------------------------- ------------ -----------
        101 Thansila
Begam                                                         2        4200

        104 Divya
Rani                                                          1        4200

        103 Rahul
Kumar                                                         1        3200
```

9. Payment Transaction Report

This query generates a payment transaction report using JOIN to display order details, customer names, payment methods, payment statuses, and payment amounts

```
SQL> SELECT o.order_id,
  2         c.first_name,
  3         c.last_name,
  4         p.payment_method,
  5         p.payment_status,
  6         p.amount
  7  FROM orders o
  8  JOIN customer c
  9  ON o.customer_id = c.customer_id
 10  JOIN payment p
 11  ON o.order_id = p.order_id
 12  ORDER BY o.order_date DESC;

  ORDER_ID FIRST_NAME
---------- --------------------------------------------------
LAST_NAME                                          PAYMENT_METHOD
-------------------------------------------------- ----------------
PAYMENT_STATUS           AMOUNT
-------------------- ----------
      1002 Priya
Devi                                               Credit Card
Success                    1800

      1002 Priya
Devi                                               Credit Card
Success                    1800

  ORDER_ID FIRST_NAME
---------- --------------------------------------------------
LAST_NAME                                          PAYMENT_METHOD
-------------------------------------------------- ----------------
PAYMENT_STATUS           AMOUNT
-------------------- ----------


      1002 Priya
Devi                                               Credit Card
Success                    1800
```